

# Analysis 1: Full-Feature Elastic-Net Nomogram at the Hour-6 Landmark

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### 0.1 Overview

This report builds a full-feature penalised-logistic nomogram at the hour-6 landmark, establishing the statistical upper bound for predicting incident Sepsis-3 within 12 hours. Heavily-missing labs are incorporated via an informative-missingness design (median-impute plus “was it measured” indicators). Evaluation uses nested cross-validation with DeLong testing against the earlier baseline model.

### 0.2 Load Libraries

```
suppressPackageStartupMessages({  
  library(glmnet); library(pROC); library(PROC); library(rms)  
})  
set.seed(486649)
```

### 0.3 Locate Data and Set Output Directories

```
find_data <- function(f) {  
  cands <- c(file.path("../..//processed_data", f),  
             file.path("processed_data", f), f)  
  hit <- cands[file.exists(cands)][1]
```

```

  if (is.na(hit)) stop("cannot find ", f)
  hit
}
OUT <- dirname(find_data("08_landmark_h6.parquet"))
FIG <- if (dir.exists("figure")) "figure" else { dir.create("figure"); "figure" }

```

## 0.4 Load Landmark Data

```

lm6 <- as.data.frame(arrow::read_parquet(find_data("08_landmark_h6.parquet")))
cat(sprintf("Loaded landmark_h6: %d stays, onset-within-12h rate %.3f\n",
           nrow(lm6), mean(lm6$onset_within_h)))

```

```
## Loaded landmark_h6: 48829 stays, onset-within-12h rate 0.068
```

## 0.5 Feature Design

The feature matrix combines vitals (near-complete after forward-fill), severity scores (SOFA, qSOFA, SIRS), labs with informative-missingness indicators, and 6-hour trajectory slopes. Labs (70-88% missing at hour 6) and slopes (~30% missing) use median/zero imputation with a binary “measured” flag, preserving the informative nature of missingness: a drawn lactate itself marks a sicker or suspected patient.

```

vitals <- c("hr", "resp_rate", "map", "spo2", "temp_c", "gcs")
scores <- c("sofa_total", "qsofa", "sirs")
labs <- c("lactate", "creatinine", "wbc", "pf_ratio")
slopes <- c("hr_slope6", "resp_rate_slope6", "map_slope6", "spo2_slope6")

df <- data.frame(row.names = seq_len(nrow(lm6)))
for (v in c(vitals, scores)) {
  x <- lm6[[v]]; x[is.na(x)] <- median(x, na.rm = TRUE); df[[v]] <- x
}
for (v in labs) {
  x <- lm6[[v]]; df[[paste0(v, "_msg")]] <- as.integer(is.na(x))
  x[is.na(x)] <- median(x, na.rm = TRUE); df[[v]] <- x
}
for (v in slopes) {
  x <- lm6[[v]]; df[[paste0(v, "_msg")]] <- as.integer(is.na(x))
  x[is.na(x)] <- 0; df[[v]] <- x
}
df$age <- lm6$age
df$female <- as.integer(lm6$gender == "F")
y <- lm6$onset_within_h
X <- as.matrix(df)
cat(sprintf("Design matrix: %d features\n", ncol(X)))

```

```
## Design matrix: 27 features
```

## 0.6 Outer 5-Fold CV with Inner Lambda Selection

Nested cross-validation provides honest evaluation: the outer loop splits stays into 5 folds and the inner loop (inside `cv.glmnet`) selects the optimal regularisation parameter on each training fold.

```
folds <- sample(rep(1:5, length.out = nrow(X)))
oos_en <- numeric(nrow(X))
ALPHA <- 0.5
for (k in 1:5) {
  tr <- folds != k
  cvfit <- cv.glmnet(X[tr, ], y[tr], family = "binomial",
                    alpha = ALPHA, nfolds = 5, standardize = TRUE)
  oos_en[!tr] <- as.numeric(predict(cvfit, X[!tr, ],
                                   s = "lambda.min", type = "response"))
}
```

## 0.7 Reconstruct Baseline Models

The baseline GLMs from script 09 are re-fit on the same outer folds for a fair DeLong comparison, using the imputed matrix so all models share identical rows.

```
gdat <- cbind(df, y = y)
cvauc_glm <- function(form, data) {
  p <- numeric(nrow(data))
  for (k in 1:5) { tr <- folds != k
    m <- glm(form, data[tr, ], family = binomial)
    p[!tr] <- predict(m, data[!tr, ], type = "response") }
  p
}
p_static <- cvauc_glm(y ~ age + sofa_total + qsofa + sirs +
  hr + resp_rate + map + spo2 + temp_c, gdat)
p_traj <- cvauc_glm(y ~ age + sofa_total + qsofa + sirs +
  hr + resp_rate + map + spo2 + temp_c +
  hr_slope6 + resp_rate_slope6 + map_slope6 + spo2_slope6,
  gdat)
```

## 0.8 Compute Metrics

```
auc_of <- function(p) as.numeric(auc(roc(y, p, quiet = TRUE, direction = "<")))
prauc_of <- function(p) pr.curve(scores.class0 = p[y==1],
                                scores.class1 = p[y==0])$auc.integral
brier <- function(p) mean((p - y)^2)

roc_en <- roc(y, oos_en, quiet = TRUE, direction = "<")
roc_trj <- roc(y, p_traj, quiet = TRUE, direction = "<")
dl_vs_traj <- roc.test(roc_en, roc_trj, method = "delong")$p.value

metrics <- data.frame(
  model = c("Baseline static (glm)", "Baseline + trajectory slopes (glm, 0.755 AUROC)",
            "Analysis1 full-feature elastic net"),
```

```

auroc = c(auc_of(p_static), auc_of(p_traj), auc_of(oos_en)),
prauc = c(prauc_of(p_static), prauc_of(p_traj), prauc_of(oos_en)),
brier = c(brier(p_static), brier(p_traj), brier(oos_en))
metrics$delong_p_vs_baseline_traj <- c(NA, NA, dl_vs_traj)
metrics[, -1] <- round(metrics[, -1], 4)
print(metrics, row.names = FALSE)

##                                model  auroc  prauc  brier
##                                Baseline static (glm) 0.7488 0.1735 0.0600
## Baseline + trajectory slopes (glm, 0.755 AUROC) 0.7546 0.1793 0.0598
##                                Analysis1 full-feature elastic net 0.7752 0.2064 0.0587
## delong_p_vs_baseline_traj
##                                NA
##                                NA
##                                0

write.csv(metrics, file.path(OUT, "10_analysis1_metrics.csv"), row.names = FALSE)

```

## 0.9 Final Model Coefficients

The elastic-net model is fit on all data to extract penalised coefficients and odds ratios. Only features with non-zero coefficients are retained.

```

cvfull <- cv.glmnet(X, y, family = "binomial", alpha = ALPHA,
                    nolds = 5, standardize = TRUE)
co <- as.matrix(coef(cvfull, s = "lambda.min"))
coefs <- data.frame(term = rownames(co), coef = as.numeric(co))
coefs$odds_ratio <- exp(coefs$coef)
coefs <- coefs[coefs$coef != 0, ]
coefs[, c("coef", "odds_ratio")] <- round(coefs[, c("coef", "odds_ratio")], 4)
write.csv(coefs, file.path(OUT, "10_analysis1_coefs.csv"), row.names = FALSE)
cat(sprintf("\nSelected features (non-zero at lambda.min): %d / %d\n",
            nrow(coefs) - 1, ncol(X)))

##
## Selected features (non-zero at lambda.min): 25 / 27

```

## 0.10 Export Out-of-Sample Predictions

Per-stay out-of-sample risk estimates feed into the Analysis 4 alarm/utility analysis downstream.

```

arrow::write_parquet(data.frame(stay_id = lm6$stay_id, onset_within_h = y,
                                risk_en = oos_en, risk_baseline_traj = p_traj),
                    file.path(OUT, "10_analysis1_oos_pred.parquet"))

```

## 0.11 Calibration Curve

```

png(file.path(FIG, "10_calibration_h6.png"), width = 1000, height = 1000, res = 200)
bins <- cut(oos_en, quantile(oos_en, seq(0, 1, 0.1)), include.lowest = TRUE)
cal <- aggregate(cbind(pred = oos_en, obs = y), list(bin = bins), mean)

```

```
plot(cal$pred, cal$obs, pch = 19, xlab = "Predicted risk",
      ylab = "Observed onset rate", main = "Calibration (out-of-sample, h6)",
      xlim = range(cal$pred, cal$obs), ylim = range(cal$pred, cal$obs))
abline(0, 1, lty = 2, col = "grey50"); lines(cal$pred, cal$obs, col = "steelblue")
dev.off()
```

```
## pdf
## 2
```

## 0.12 Nomogram

An unpenalised logistic regression (lrm) is refit on the features selected by elastic net, producing a clinician-facing nomogram.

```
sel <- setdiff(coefs$term, "(Intercept)")
nomo_df <- df[, intersect(sel, names(df)), drop = FALSE]; nomo_df$y <- y
dd <- datadist(nomo_df); options(datadist = "dd")
frm <- as.formula(paste("y ~", paste(names(nomo_df)[names(nomo_df) != "y"],
                                   collapse = " + ")))
fit_lrm <- lrm(frm, data = nomo_df, maxit = 1000)
png(file.path(FIG, "10_nomogram_h6.png"), width = 1800, height = 1400, res = 180)
plot(nomogram(fit_lrm, fun = plogis, funlabel = "Risk of sepsis within 12h"),
      cex.axis = 0.6, cex.var = 0.7)
dev.off()
```

```
## pdf
## 2
```

## 0.13 Done

```
cat("\nAnalysis 1 complete. Outputs in", OUT, "and", FIG, "\n")
```

```
##
## Analysis 1 complete. Outputs in ../../processed_data and figure
```